

for working with quantum computers) and Quantum Lab (allows online code development for quantum circuits), which are available to the programming community for testing and development. Another big player in the field of quantum computing is Microsoft. Quantum Development Kit (QDK) and Q# (Q Sharp, a quantum programming language) are some of the important contributions made by Microsoft to quantum computing.

Quantum computer simulators

Now that we have seen the contributions from different companies and organisations to quantum computing, let us discuss quantum computer simulators. We have seen that developing quantum computers is an extremely difficult task. So, how can researchers test a new quantum algorithm? Quantum simulators help us achieve this by using classical computer hardware to imitate quantum behaviour. The difficulty in simulating quantum behaviour arises from the exponential time requirement by classical computers — as the number of qubits increases, the simulation becomes slower and slower. QuEST, developed by the University of Oxford, is an example of quantum computer simulators.

The development of these simulators has also led to the recent popularisation of quantum clouds, a platform that is trending nowadays. Yes, it is true! I have seen quite a number of advertisements offering free quantum cloud time and if entrepreneurs are pouring in their resources, there ought to be some truth behind these. Some of these quantum computer simulators are based on the quantum annealing technique. I believe that in the near future a lot of startups will come up with brilliant ideas to offer more and more quantum cloud resources. An example of quantum cloud services is the IBM Quantum Experience. It allows users to graphically build quantum circuits and develop quantum programs using Qiskit, an open source quantum software development kit (see Figure 3).



Figure 3: The IBM Quantum Experience interface

Quantum programming languages

Any general discussion on quantum computers will be incomplete if we do not discuss quantum programming languages. What are the important quantum programming languages? We have already mentioned Q# (Q Sharp), a quantum programming language developed by Microsoft. It is a free and open source programming language licensed under the MIT License. This programming language allows us to use qubits in algorithms. The syntax of Q# is related to the syntax of C#, a classical programming language, also developed by Microsoft. Q# is a high-level programming language, whereas OpenQASM is an assembly language used for quantum programming. An implementation of OpenQASM is used with the IBM Quantum Experience quantum cloud computing platform. It is a free and open source programming language licensed under the Apache License. Cirq is a framework for Noisy Intermediate Scale Quantum (NISQ) computers, developed by Google. This is also free and open source software licensed under the Apache License. Cirq uses the Python programming language for developing quantum algorithms. There are a lot of other quantum programming languages, but the support by Microsoft, IBM and Google makes the above mentioned languages significant.

The role of open source

Before we wind up our discussion, it is essential to answer an important question. What is the role played by the open source community in quantum computing? Of course, quantum computing is a field where you need a gigantic budget for hardware research. But, what about the programming community? There might be a day when there are millions of quantum computers in the world, just like the classical computers we have today. The best news for the open source community is that almost every software related to quantum computing is open source. We have already seen examples of quantum programming languages, and almost all of them are free and open source. The same goes for other quantum computing resources also. Being a new technology, it is essential to have the support of a large community. I believe the developers of quantum computing feel that the open source model is the best to sustain and nurture such a developing technology. Let us hope that the year 2020 for quantum computing will be the same as the year 1950 was for classical computing. We only had two digital computers, UNIVAC and EDSAC, in 1950. Look at the computing power we have today. Hopefully, we will witness a similar revolution in quantum computing in the near future. 

By: Deepu Benson

The author is a free software enthusiast whose area of interest is theoretical computer science. The open source tools of his choice include ns-2 and ns-3. He maintains a technical blog at www.computingforbeginners.blogspot.in.